

# Atlas of halving: what one refinement step does, and what emerges, in each dimension

**Purpose, 2026-09-27 (user direction: a collective understanding of the halving of the lattice and what emerges in each case and dimension, as the mid-term goal).** This note is the shared map. Each cell says what one halving does, what it leaves behind, and what survives the limit, with the status of the statement and the note that holds it. It adds no new theorem; it fixes one vocabulary so that the agents working here (Claude, GPT-6 Astra, Fable reviewers) and the user fill the same cells. Open cells are listed in § 5.

Status labels: **proved** (theorem in a repository note, reviewed or not as marked), **exact** (closed-form identity), **formal** (derivation assuming an unproved estimate), **known** (established literature, cited), **open**.

## 1. The operation

**Halving one direction**  $a_1 \mapsto a_1/2$  of a lattice with plaquette heat times  $t_{\mu\nu} = \lambda_D a_\mu a_\nu / \prod_{\rho \neq \mu, \nu} a_\rho$  (series/parallel note, eq. (1), small-field regime) splits exactly into two moves.

- **Series move.** A face containing the refined direction is cut in two; its heat time halves. Integrating the cutting edge is a convolution of the two half-weights: character coefficients multiply. For heat-kernel weights it closes exactly in every dimension.
- **Parallel move.** A face transverse to the refined direction gets a new copy in the mid-plane, both at doubled heat time. Integrating the mid-plane leaves the factor  $\Psi$ : a  $(D-1)$ -dimensional gauge theory whose edges carry Brownian-bridge laws fixed by the coarse field. Pointwise products of weights add their logarithms; the heat kernel is closed under this only up to vortex terms (zero-spacing note, Proposition 1).
- **Count.** A halving produces parallel moves in  $\binom{D-1}{2}$  transverse planes. An isotropic step is  $D$  directional halvings; the isotropic heat time scales as  $t \propto \lambda_D a^{4-D}$ .
- **Time-only halving** (Euclidean time at fixed spatial lattice) sends the transverse heat times to infinity; the Hamiltonian limit needs exponential (Wilson-type) magnetic weights (zero-spacing note, § 1).

**Newton's halving** is the one-dimensional case: inserting a time  $t_{2.6}$  in a cell is a series move (refinement note, § 2). A physical record at the inserted time is a new variable attached to the body, the analogue of a parallel insertion. Forgetting a Gaussian record is an exact parallel move with defect proportional to  $\hbar^2$ , the identity for  $\hbar = 0$  (record as parallel move).

### 1b. Cuts at any position (user remarks, 2026-09-27/28)

Halving is the easiest cut; a cut at any fraction  $s \in (0, 1)$  of a cell performs the same two moves, with  $s$  as a parameter.

- **Series move.** The cut face splits into heat times  $st$  and  $(1-s)t$ ; the convolution returns  $t$  exactly, for heat-kernel weights, in every dimension.

- **Parallel move.** With trapezoid (dual-length) face weights every transverse face keeps heat time  $2t$ , and the inserted edges carry the bridge law at fraction  $s$ , variance  $s(1-s)t_e$ . The free-field defect scales exactly by  $4s(1-s)$  (Corollary 2<sub>s</sub>), and the  $U(1)$  Theorem 5 holds for every  $s$  (Corollary 5<sub>s</sub> there): halving is the largest single step, and an off-centre cut is gentler and shrinks the mesh less.
- **Newton.** One cut spends exactly  $3s(1-s)K_\tau$  of the cell action  $K_\tau = F^2\tau^3/(24m)$ , the Cameron–Martin energy of the Schauder hat it inserts; the shares add over any cut sequence and exhaust  $K_\tau$  iff the mesh vanishes (cut-measure note). The Galileo action is an additive measure on the cut process, with the nested additivity of the Lévy–Ciesielski construction of the Brownian bridge. With a mark floor  $\kappa$ , marks at any finite set of cuts distinguish the force with squared statistical distance exactly the spent action over  $\kappa$  (Proposition 7 there), so the cut measure and the record measure coincide.
- **Windings and images.** For  $U(1)$  a cut’s bridge law is a positive winding mixture: its weights are  $s$ -independent and the shifts  $2\pi(1-s)W$  form  $\mathbb{Z}_2$  only at halving and  $\mathbb{Z}_q$  at  $s = p/q$ . For every compact simply connected group the midpoint expectation of every character is exact, a sum over weights of coroot image sums (centre note, Theorems 1 and 1’). Its signed Weyl-polynomial amplitudes describe relative Cartan images; positivity of a mixture and homotopy winding sectors in the group are additional interpretations unsupported by that expansion.
- **Which part of the centre a cut reaches.** A cut at  $p/q$  reaches exactly the subgroup  $\mathbb{Z}_{\gcd(q,N)}$  of the centre of  $SU(N)$  (its Corollary 3). So the  $SU(2)$  halving carries Dirac’s belt trick, the image  $n$  acting by  $(-1)^n$  on half-integer spins, while dyadic refinement reaches only its identity for  $SU(3)$  and a trisection carries the triality. Flipping one mid-edge by a central element inserts thin centre flux into its incident faces. Open: whether this ties refinement to centre vortices, and whether triadic refinement suits  $SU(3)$  better. Astra’s Round 5B Part A referee is complete (2026-09-28): the exact identities and torsion criterion are accepted, with scope corrections.
- **The rod.** The stick of Zhuangzi 33, 一尺之捶, 日取其半, 萬世不竭, halved only in its remaining piece, spends  $\frac{6}{7}K_\tau$  and converges to a point, the Mohist 端 of Canon B; refinement, which spends all of  $K_\tau$ , cuts every piece again (cut-measure note, § 3). The parallel with Zeno’s dichotomy is recorded as convergence.
- **The frame.** A cut at an arbitrary, even irrational, position is the geometers’ cut of Book I’s closing scholium. The questions for each cell become: does the limit depend on the cut sequence (for Newton, no, proved in the cut-measure note; for the free field, expected, a convergence question for the discrete-exterior-calculus Hodge star that the trapezoid weights are, see Corollary 2<sub>s</sub>), and does the universal part survive when  $s$  varies. For cell 4 it does, conditionally: for shape-regular schedules under the matching hypotheses of the four-dimensional note (its § 7), the coefficient per logarithm of the physical scale is  $2b_0$  whatever the cut positions. An earlier guess here,  $\log(1/s)$  per cut, follows one daughter only and omits the other’s dual-volume weight.

## 2. The atlas by dimension

| $1 + n$           | Series move                                                                                                 | Parallel planes per halving | What one step leaves                                                                                                                                                                     | What survives the limit                                                                                                           | Status and notes                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| $0 + 0$           | none (a single weight)                                                                                      | none                        | the weight $k_t(U)$ itself; a single integral with an $e^{-c/\hbar}$ structure                                                                                                           | a large- $N$ transition (Gross–Witten)                                                                                            | known; dimension ladder                                                  |
| $1 + 0$<br>Newton | exact after one cubic counterterm<br>$-F^2\hbar^3/(24M)$ ;<br>the defect $-F^2uv\hbar/(8M)$ is a coboundary | none                        | a scalar                                                                                                                                                                                 | unobserved: the exact propagator at every $\hbar$ ; recorded: a floor only if joint determinacy fails                             | proved; refinement note, fifth postulate                                 |
| $1 + 1$           | exact (heat-kernel convolution)                                                                             | 0                           | nothing for heat kernels; for any action, $a^{-2}(1 - \hat{c}_R)$                                                                                                                        | a conjugation-invariant Lévy exponent $\psi(R)$ ; Yang–Mills iff Lindeberg; circle spectrum $\hbar cL\psi(R)$ ; no local particle | proved (Theorem 2 of the zero-spacing note); known (Lévy)                |
| $1 + 2$           | exact                                                                                                       | 1                           | free: an irrelevant quadratic form of relative size $a^2K^2/16 + (Ga)^2/8$ ; $U(1)$ : vortices of density $e^{-\pi^2/(8t)}$ ; $SU(N)$ : coupling shifts $O(t)$ , curvature term explicit | free photon for $U(1)$ at fixed $\lambda_3$ (every lattice gap dies); conjectured gap $C_3\hbar c\lambda_3$ for $SU(N)$           | proved for free and $U(1)$ ; formal for $SU(N)$ (Proposition 7); see § 3 |

| $1 + n$           | Series move | Parallel planes per halving | What one step leaves                                                                                        | What survives the limit                                                                                                                                                                        | Status and notes                                                                           |
|-------------------|-------------|-----------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| $1 + 3$           | exact       | 3                           | $t = g^2$ fixed: coupling shifts that recur each step; large fields suppressed only as $(a\Lambda)^{2b_0c}$ | running coupling, $\Lambda = \mu e^{-1/(2b_0g^2)}$ ; small instantons die $(11N/3 > 4)$ ; dislocations need $c > 2/b_0$ ( $\frac{6}{11}$ of an instanton for $SU(2)$ ); $U(1)$ : Coulomb phase | known (running, 6/11 criterion, Driver); the measure-level route fails for $U(1)$ (proved) |
| $1 + n, n \geq 4$ | exact       | $\binom{n}{2}$              | $t$ grows as $a \rightarrow 0$                                                                              | no small-field regime; a phase transition, no established continuum limit                                                                                                                      | known (Creutz for $SU(2)$ in 5D)                                                           |

The same heat time  $t = \hbar g_{\text{cl}}^2 a^{4-D}$  controls the column “what one step leaves” . For  $D < 4$  refining and taking  $\hbar \rightarrow 0$  push it the same way; in mechanics refining makes each cell more quantum (dimension ladder, § 2).

### 3. The atlas for $1 + 2$ by group

| Group                   | Isolated cube | Full mid-plane                                           | One step, measure level | Iteration and limit                                                                                 |
|-------------------------|---------------|----------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------|
| free ( $\mathbb{R}^n$ ) | exact         | exact defect (Proposition 2 of the series/parallel note) | Gaussian, exact         | blocked actions converge to a Gaussian fixed point (known, Bell–Wilson; for this blocking asserted) |

| Group   | Isolated cube                                                                                                                                                                                 | Full mid-plane                                                                                                                                                                                                                                                                                                              | One step,<br>measure level                                                                                                                                                                                                         | Iteration and<br>limit                                                    |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| $U(1)$  | exact<br>(Proposition 3)                                                                                                                                                                      | exact charge<br>form, vortex<br>bound<br>(Proposition 4,<br>Corollary)                                                                                                                                                                                                                                                      | one step<br>equals the free<br>step up to<br>density $e^{-\pi^2/(8t)}$<br>(Theorem 5,<br>refereed);<br>whole measure<br>within TV<br>$2(L/a)^3 e^{-\pi^2/(6\lambda_3 a)}$<br>of the<br>monopole-free<br>part<br>(monopole<br>note) | free photon<br>(Gross 1983,<br>known)                                     |
| $SU(2)$ | midpoint<br>characters<br>exact for every<br>spin; spin- $\frac{1}{2}$<br>and spin-1<br>cube sectors<br>exact, with<br>matrix image<br>bounds for<br>spin 1 (closed<br>form,<br>Theorems 1–4) | formal order- $t$<br>expansion: cut<br>shifts positive,<br>transverse<br>negative; other<br>local operators<br>have<br>dimension at<br>least six;<br>explicit torus<br>winding term                                                                                                                                         | formal;<br>one-step<br>normalized<br>small-field<br>bound against<br>a covariant<br>reference<br>proved<br>(small-field<br>§ § 13–14);<br>iteration open                                                                           | open                                                                      |
| $SU(3)$ | curvature term<br>explicit,<br>softening<br>$t_j  X_j ^2/512$<br>(group-general<br>form of<br>Proposition 6)                                                                                  | small-field<br>one-step<br>normalized<br>bounds<br>transcribed<br>from $SU(2)$ , $\mathbb{Z}_3$<br>sectors, bridge<br>tail by Li–Yau<br>(small-field<br>§ 16, refereed<br>by Astra with<br>corrections);<br>bulk order- $t$<br>coefficients $\frac{3}{2}$<br>times $SU(2)$ 's<br>(order- $t$ § 7b);<br>large fields<br>open | formal                                                                                                                                                                                                                             | open; the gap<br>$C_3 \hbar c \lambda_3$ is the<br>infrared<br>obligation |

## 4. What emerges, read across the atlas

Why ultraviolet halvings bear on an infrared gap is explained, for the paper, in the didactic note.

- **Series moves never produce anything new.** In every dimension they close exactly for heat-kernel weights, and in mechanics after one scalar counterterm. The one-dimensional Newton problem and the two-dimensional gauge theory are all series.
- **Parallel moves carry everything that renormalizes.** Their number,  $\binom{D-1}{2}$ , is zero exactly where the theory is exactly soluble.
- **Additivity is universal; locality is one-dimensional** (2026-09-28). Integrating out a Gaussian step leaves the Schur complement of the action (the Dirichlet principle), and successive Schur complements compose exactly ((13) of the four-dimensional note); in Newton's cell this is Theorem 2 of the cut-measure note. Only one dimension grants locality: the minimizer between two cut points is the chord, so the blocked action is again nearest-neighbour and the series move closes, as in  $1 + 1$ . With transverse planes the Schur complement is non-local, and its local truncation error is the parallel defect ((5) and Corollary 2<sub>s</sub> of the series/parallel note).
- **The exponential ladder.** Each compact correction left by a step is  $e^{-\chi/t}$ , with dimensionless  $\chi > 0$  (vortices, monopoles, dislocations). In  $D < 4$  it is summable per physical volume at fixed coupling, so compact effects die; in  $D = 4$  the running turns it into a power of  $a$ , so survival is a threshold condition; in  $D > 4$  there is no small  $t$ . In  $1 + 3$  at one loop,  $a\Lambda = e^{-1/(2b_0t)}$  with  $t = g^2(a) = \hbar g_{\text{cl}}^2(a)$ , so  $e^{-\chi/t} = (a\Lambda)^{2b_0\chi}$  and  $2b_0\chi$  is the scaling dimension of the correction. This is the bookkeeping of 't Hooft's infrared renormalons, whose Borel-plane positions are fixed by  $b_0$  and an operator dimension; the per-volume threshold  $2b_0\chi = 4$  sits at the gluon-condensate renormalon ('t Hooft, Erice 1977, publ. 1979; metadata, renormalon positions quoted from memory). The generated gap is itself on the ladder conditionally: if  $E_{\text{gap}}$  is of order  $\hbar c\Lambda$ , its dimensionless lattice gap  $aE_{\text{gap}}/(\hbar c)$  is of order  $e^{-1/(2b_0g^2)}$ , the member with  $2b_0\chi = 1$ . The bridge image terms sit far above the per-volume threshold  $2b_0\chi = 4$ :  $2b_0\chi = 11N/3$  for root images ( $\chi = 8\pi^2$ ) and 33 for the  $SU(3)$  centre images of a trisection ( $\chi = 24\pi^2$ , centre note) (2026-09-28). These are formal powers from near-identity image exponents; normalized bounds uniform in boundary data and under iteration remain to prove.
- **What survives is trajectory-dependent.** Compact  $U(1)$  in  $1 + 2$  keeps a gap along  $\lambda_3 a \simeq c_0/(2 \log(1/a))$ ,  $c_0 \approx 4.99$  the monopole exponent, where the monopole density per physical volume diverges, and loses it at fixed  $\lambda_3$  (Göpfert–Mack versus Gross).
- **The action floor survives what removes the mass gap** (user observation, 2026-09-28). With  $t = \lambda_3 a = \hbar g_{\text{cl}}^2 a$  the compact corrections of a step,  $e^{-c/(\hbar g_{\text{cl}}^2 a)}$ , vanish as  $a \rightarrow 0$  at fixed coupling and as  $\hbar \rightarrow 0$  at fixed  $g_{\text{cl}}$  and  $a$ , while the  $w = 0$  Gaussian term of (9) of the series/parallel note, of width set by  $\hbar$ , remains. Both  $U(1)$  continuum limits on record are Gaussian: Gross's photon, with field-strength covariance  $\hbar g_{\text{cl}}^2 K_0$ , and Göpfert–Mack's free massive scalar, canonically normalized along a trajectory on which  $\lambda_3$  diverges. The refinement removes the lattice mass gap and keeps the unit of action, and  $\hbar$  was first measured in this gapless theory in  $1 + 3$ , cavity radiation (Planck's talk of 14 December 1900, written up as Ann. Phys. 309, 553 (1901), through the resonators' energy elements; the field-mode reading is Debye 1910, metadata; a finite cavity keeps the box gap  $2\pi\hbar c/L$ ).

Two gaps of different kinds: a mass gap, generated and trajectory-dependent, and an action floor, supplied in  $e^{-S/\hbar}$ , that every cell of every limit keeps (each cut weighted by  $e^{-\text{share}/\hbar}$ , cut-measure Prop. 4). In G07' s terms both are unit multiples of the fixed constants; the floor is supplied, while a generated gap needs two scale-free structures that do not commute (G07 § 1; G08).

• **Constants that emerge.** Mechanics: one scalar counterterm per cell, and one action constant if joint determinacy is denied (Theorems B and B' of the fifth-postulate note). 1 + 1: the Lévy exponent (one coupling for Yang–Mills). 1 + 2: one coupling, shifted summably. 1 + 3: one scale  $\Lambda$  by dimensional transmutation.

## 5. Open cells

1.  $SU(2)$  cube: filled through  $J = 1$  (Theorem 4, exact diagonal entries, image bounds and cube sector);  $J > 1$  entries and control of the full spin sum remain open.
2.  $SU(2)$  full mid-plane in 1 + 2: order- $t$  calculation refereed as formal; small-field note § § 12–14, Rounds 14–16, refereed by Claude: a covariant Gaussian reference with ' t Hooft flux sectors, and normalized weak  $P(1/2 - \delta)$  against it for one barriered step, uniformly in plane size, for equal and unequal boundary layers with the cut faces budgeted (saddle determinant, Laplace remainder and amplitudes included; remainder densities face-local in size). Rounds 17–24 (§ § 17–24, refereed): the pair kernel (89), the link-coordinate form of (27), and the separated third response (93) hold on the barriered chart with explicit constants, uniformly in plane size; all-order response identities and a barrier-jet obstruction (§ 23); an undifferentiated small-field/large-field split with joint bad-set rarity  $p^{|H|/16}$  (§ 24); the error exponent  $\alpha' = \frac{1}{2} - 3\delta$ ,  $\delta < \frac{1}{10}$ , absorbing the proved bare scales (§ 25); the full response through third order, contacts included (§ 26, refereed). § 27 gives the three-bad-component tree bound (120), with rarity  $p^{M/64}$  and decay  $\frac{1}{2} \log(12/5)$ ; § 28 gives a factorizing reference, uniform convexity and rarity along real decoupling paths, and the one-connector bound (128) (both written, unrefereed). § 29 controls arbitrary connector moments and connected growth, and gives a uniform complex  $\ell^1$  source ball. § 30 proves the anchored two-connector tree (135) and its summed spatial reserve (140), with the earlier gradient bound retained after a proof repair. § 31 proves all-order anchored attachments (142) and a zero-free uniform polydisc at the exact Gaussian, unbarriered endpoint (Sol/Astra, written). § 32 isolates a pointwise complex-logarithm failure and proves the direct all-order density norm (148) with radial barriers in a factorized reference (2026-09-30, Sol/Astra, refereed). § 33 gives a convergent nonlinear interaction expansion with retained barriers, uniform weak-recoupling polydisc (154) and fixed spatial reserve (Sol/Astra, refereed). Next: mark the bad observable and retain rarity in its integrated density norm; recoupling to one remains separate. Then dressed activities (112) and boundary polydiscs (113), yielding (94); then target-covering paths, curvature conversion, full-integral comparison, perturbed-action stability and iteration. The fixed-frame transport-value comparison remains open; Proposition 4' s rejection stands.
3. Stability of Hypothesis  $P(\alpha)$  under iteration. For  $U(1)$  in 1 + 2, reduced to exact Gaussian blocking (Corollary 2' of the monopole note); open for  $SU(N)$ .
4. 1 + 3: composition, Thms 1–4 refereed by Claude; § 7 gives generated cubic/quartic kernels, decay, finite one-step subtracted matching with explicit constants and uniform iteration hypotheses (Round 10, formal, unrefereed); G1 refereed. Open: its (14), uniform subtracted cubic/quartic contractions;

- sublinear endpoint matching and remainder sum give the average  $-2b_0 \log 2$  (conditional Theorem 4; one-step shifts refereed).
5. Time-only halving. Structural reading, 2026-09-27, of known results. At fixed spatial lattice the electric faces make the series moves: heat-kernel electric weights are a semigroup in the time step, so they compose exactly. The magnetic faces make the parallel moves: with exponential weights  $e^{-a_0 V}$  they close exactly in their own family, since products of such weights add the exponents. The whole defect of a time halving is the non-commutation of the two families, and it vanishes as  $a_0 \rightarrow 0$  by the Trotter product formula (Trotter 1959; Chernoff 1968; metadata; strong convergence for the Laplacian on  $G^E$  plus a bounded magnetic potential). This is why the Hamiltonian limit (Kogut and Susskind 1975; transfer matrix: Creutz 1977; metadata) exists at every fixed spatial lattice, while spatial halving leaves the non-closing factor  $\Psi$ . Open: an operator-norm rate, which needs domain estimates for the commutator of the Laplacian with  $V$ .
  6. Filled 2026-09-27 for Gaussian records (record as parallel move): forgetting a record convolves momentum with variance  $\hbar^2/(4\sigma^2)$ ; precisions add in parallel; refinement converges iff  $\sum \sigma_j^{-2} < \infty$ ; identity for  $\hbar = 0$ . Complete-record tests, § § 6–7 (2026-09-29, written, unrefereed): the repaired two-pointer model obeys the area floor iff its readout error/kick determinant remains at least  $\kappa^2$  conditional on side records; a noisy kick monitor breaks a marginally saturated repair. Adaptive affinity is exact in the branchwise information (41); monitor replacement has cost (46). All refinements, § § 3b–3d construct the harmonic curve across unequal, nonnested partitions and arbitrary start/end kick mixtures, following the Lean norm obstacle (Sol/Astra, checked). The RG/’ t Hooft comparison separates curve universality, dimensional scale and action normalization. Exact retained weights define the RG transformation; their overall action factor rescales the quantum denominator as in (6q), with  $c, \hbar, G$  kept explicit. Complete-record comparisons retain the conditional noise budget (49). Hamiltonian memory, § 8 realizes the monitor cost and proves terminal posterior closure from the joint initial covariance restriction; linear blocking preserves it for the same body. § § 8.4–8.6 give a genuine two-time physical copy refinement with an explicit relative kick and joint-law defect (62), and a partition-uniform complete-record information bound (63), even at zero action (Sol/Astra, written). § § 8.7–8.9 retain every copy kick in associative Gaussian descriptors and prove a partition-independent projected terminal law with the explicit mesh rate (73). The three-cell residue (69) has rank two; the prescribed pair plus one independent additive scalar cannot reproduce it (2026-09-30, Sol/Astra, refereed). § 8.10 includes physical adaptive gains and their conjugate recoil, with complete-record posterior closure (76) and uniform continuity (77). § § 8.11–8.14 construct a continuous body-record law for the fixed physical weighted-record policy, with path-Cauchy estimate (89), partition independence, body curve (90) and posterior closure under the entire declared limiting record (92) (Sol/Astra, refereed). The limiting evolution (96) and smeared controller recoil (98) use the same finite pointer increments. § 8.15 disproves extension to unrestricted nonlinear classical body copies (103); the quantum reference supplies a curvature term (104) and an explicit Wigner-negativity witness (105). Continuity (91) also survives zero action. Next: limiting momentum/record equations and quadratic covariation. Adaptive timing, extra readouts and the independent positive scale remain open.
  7. The infrared: from ultraviolet control in  $1 + 2$  to  $C_3 > 0$ .
  8. Cuts at any position (§ 1b), open rows: the free-field independence of the limit

from the cut sequence (a convergence question for the discrete-exterior-calculus Hodge star on nested tensor grids); for  $SU(3)$ , whether triadic refinement, the only kind whose image terms reach the centre (centre note), organizes the large-field terms better than dyadic refinement.

## **6. Consequence for STATE**

STATE carries the mid-term goal and points here. Each result that fills a cell updates the cell in this note and nothing else; the notes that prove results remain the record.